

Cloud Computing Applications and Services

(Aplicações e Serviços de Computação em Nuvem)

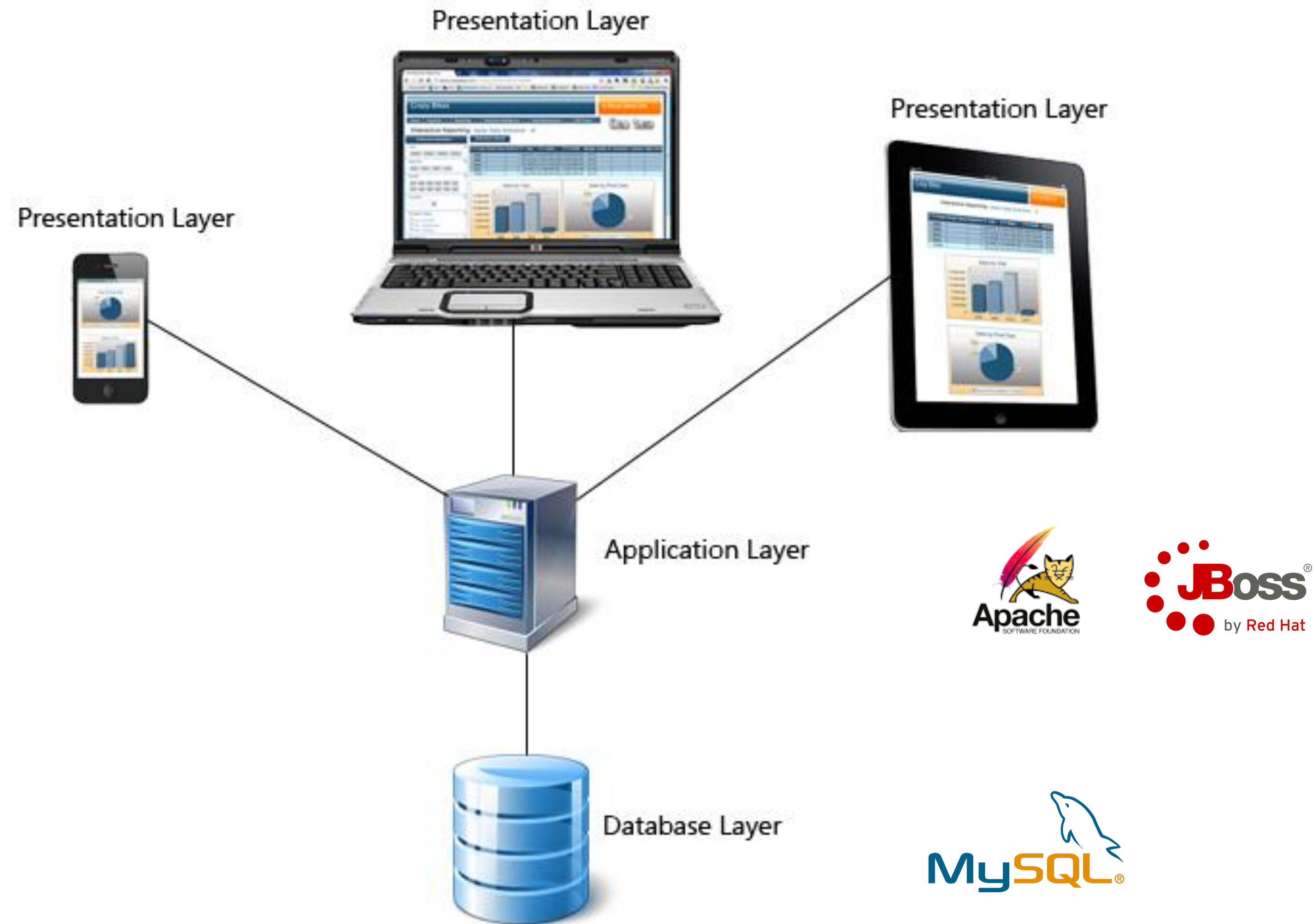
Introduction

University of Minho

2026-2027

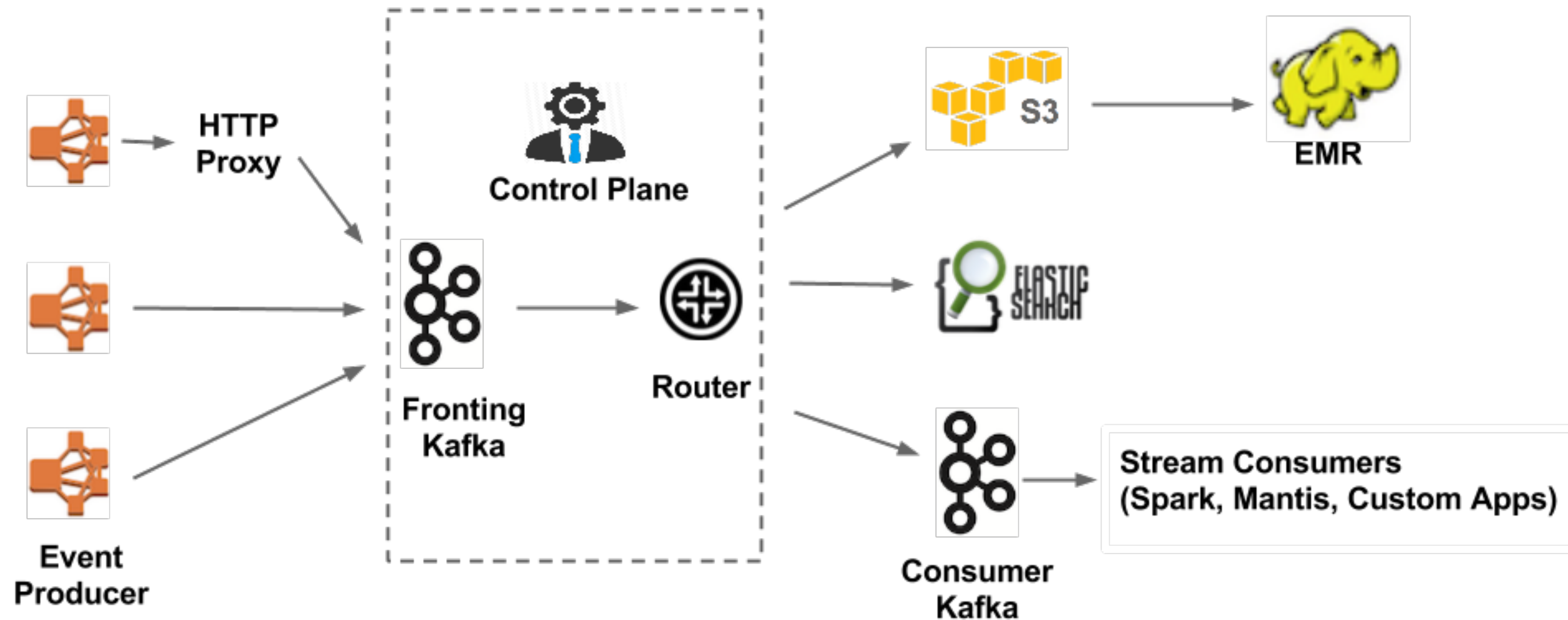


A “Simple” Application



Complex Applications

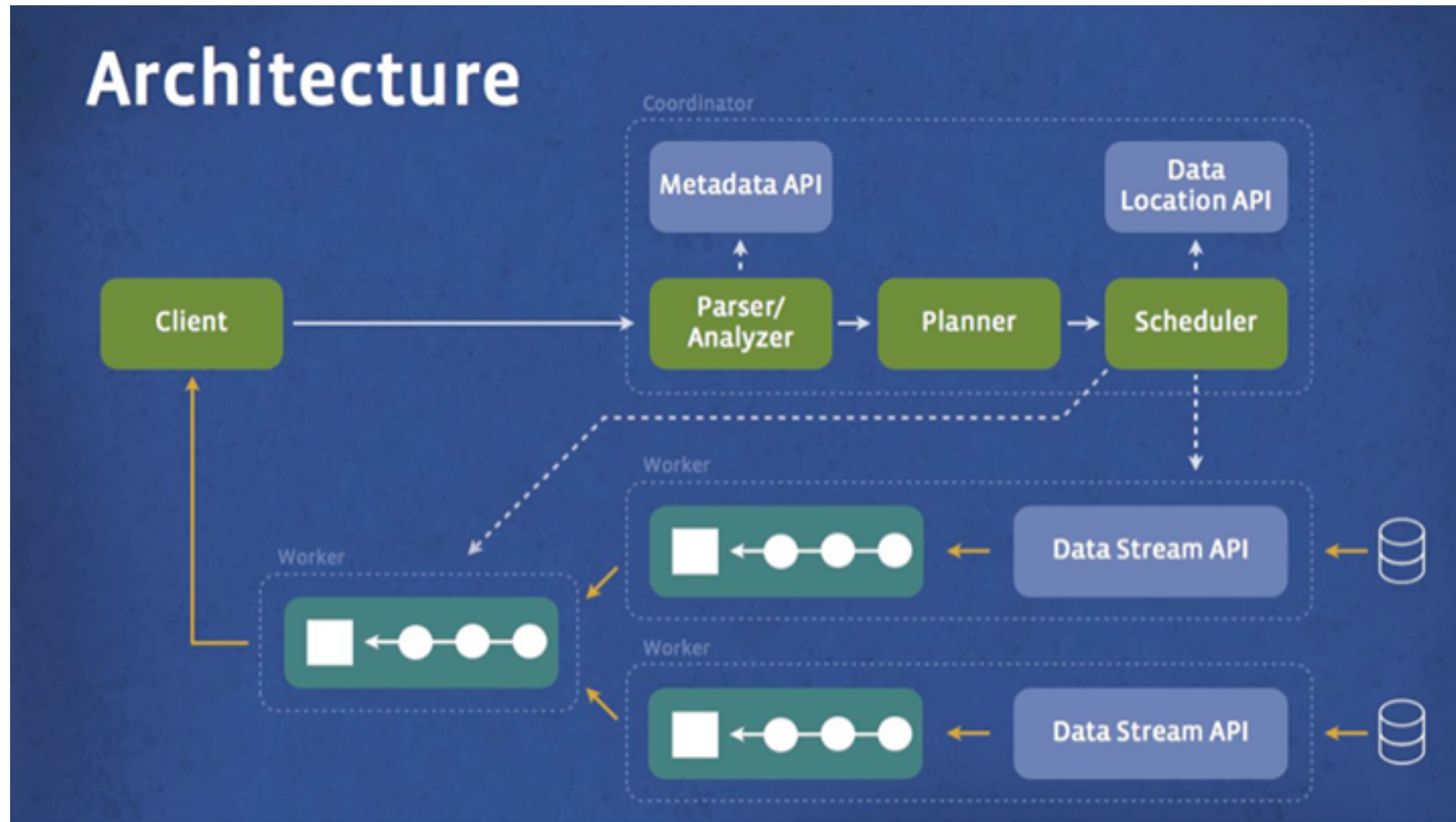
Netflix – Keystone Data Pipeline



Source: <https://medium.com/netflix-techblog/evolution-of-the-netflix-data-pipeline-da246ca36905>

Complex Applications

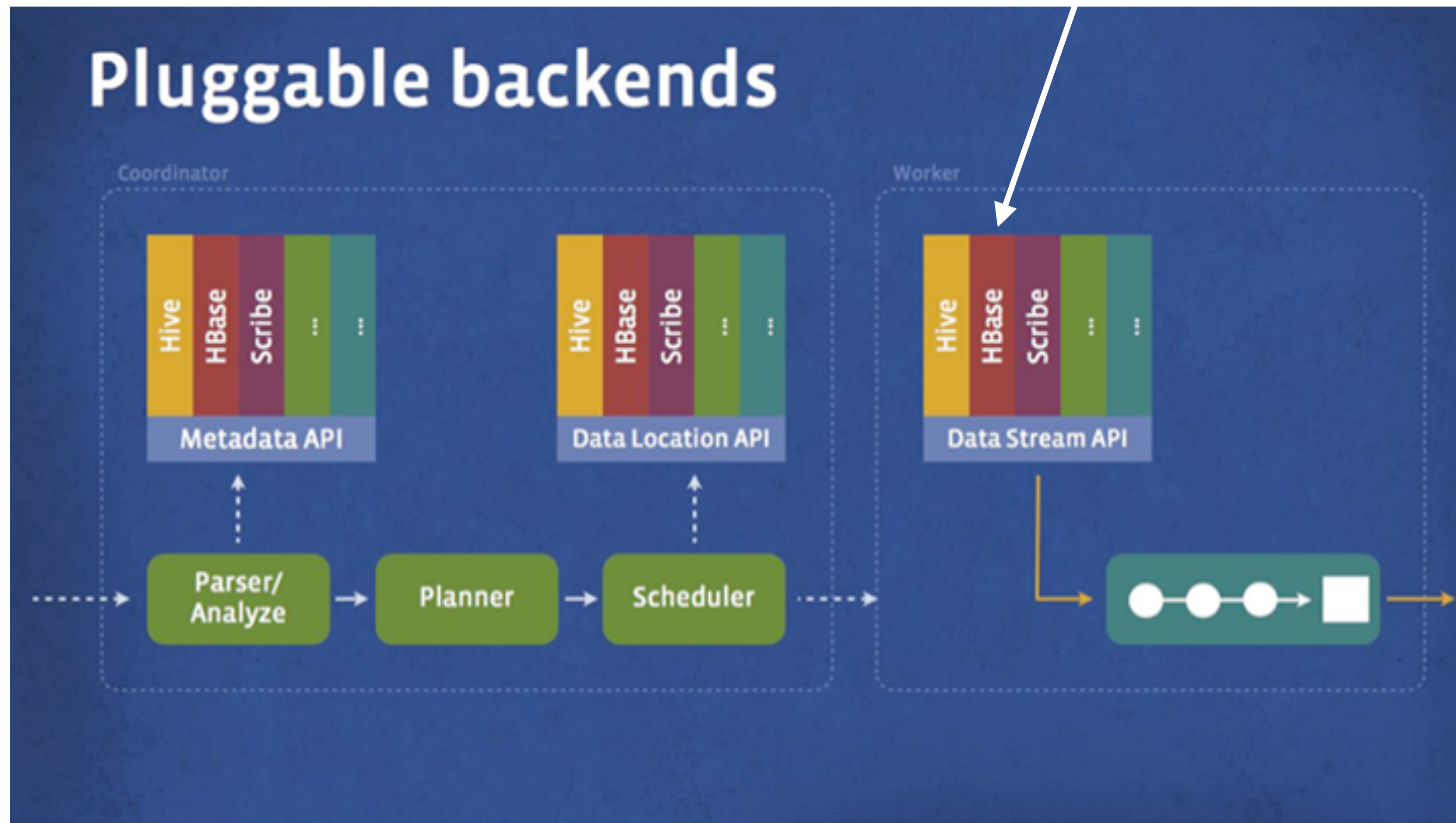
Facebook – Presto SQL Engine



Source: <https://www.facebook.com/notes/facebook-engineering/presto-interacting-with-petabytes-of-data-at-facebook/10151786197628920/?s=keen-io>

Complex Applications

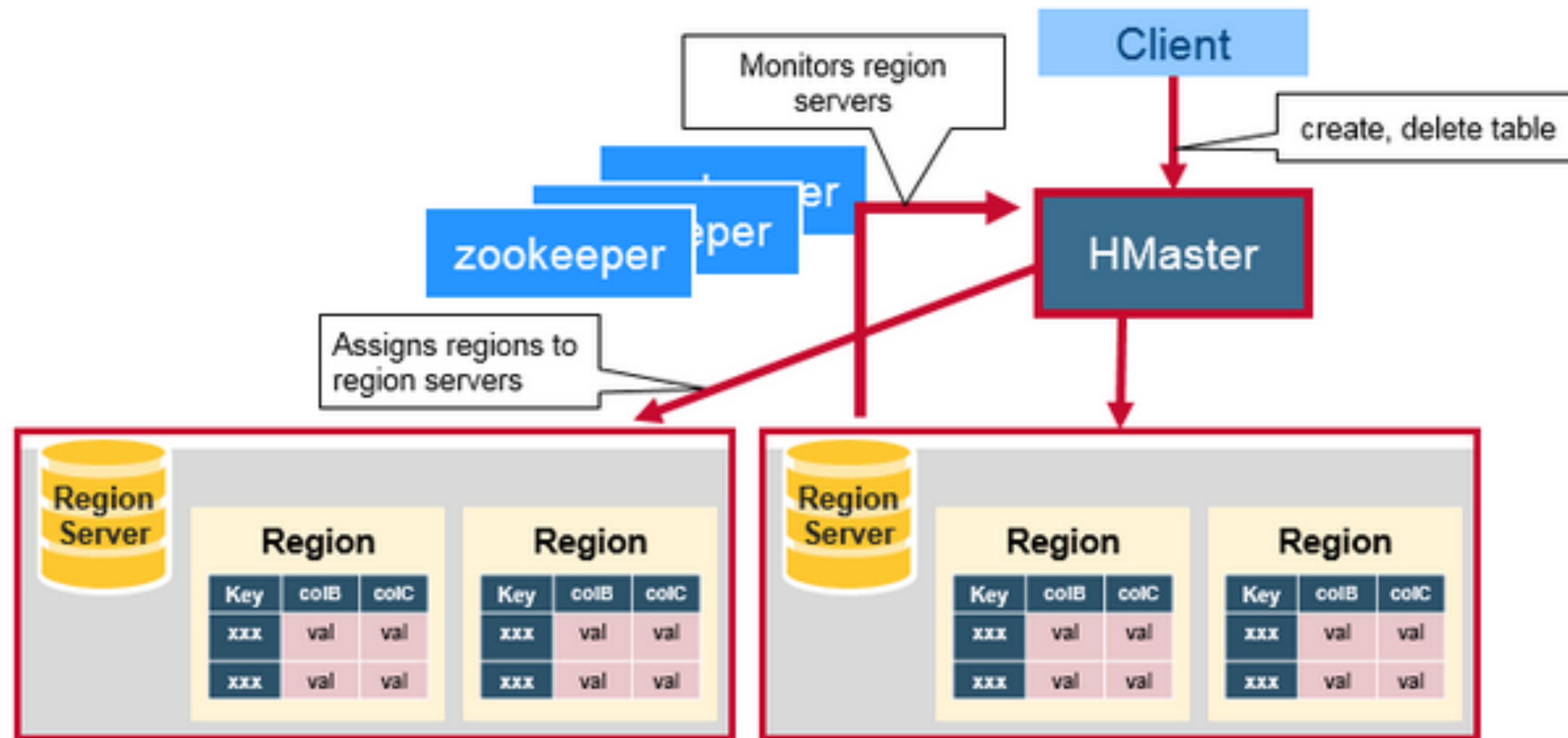
Facebook – Presto Backends



Source: <https://www.facebook.com/notes/facebook-engineering/presto-interacting-with-petabytes-of-data-at-facebook/10151786197628920/?s=keen-io>

Complex Applications

HBase



Complex Infrastructures



Challenges

● Deployment

- Applications have multiple components with different requirements (e.g., hardware, libraries)
- Heterogeneous hardware, software, and services (e.g., cloud computing services, HPC services)



● Management & Configuration

- Specific to each software/hardware component
- Optimal configurations will change over time...

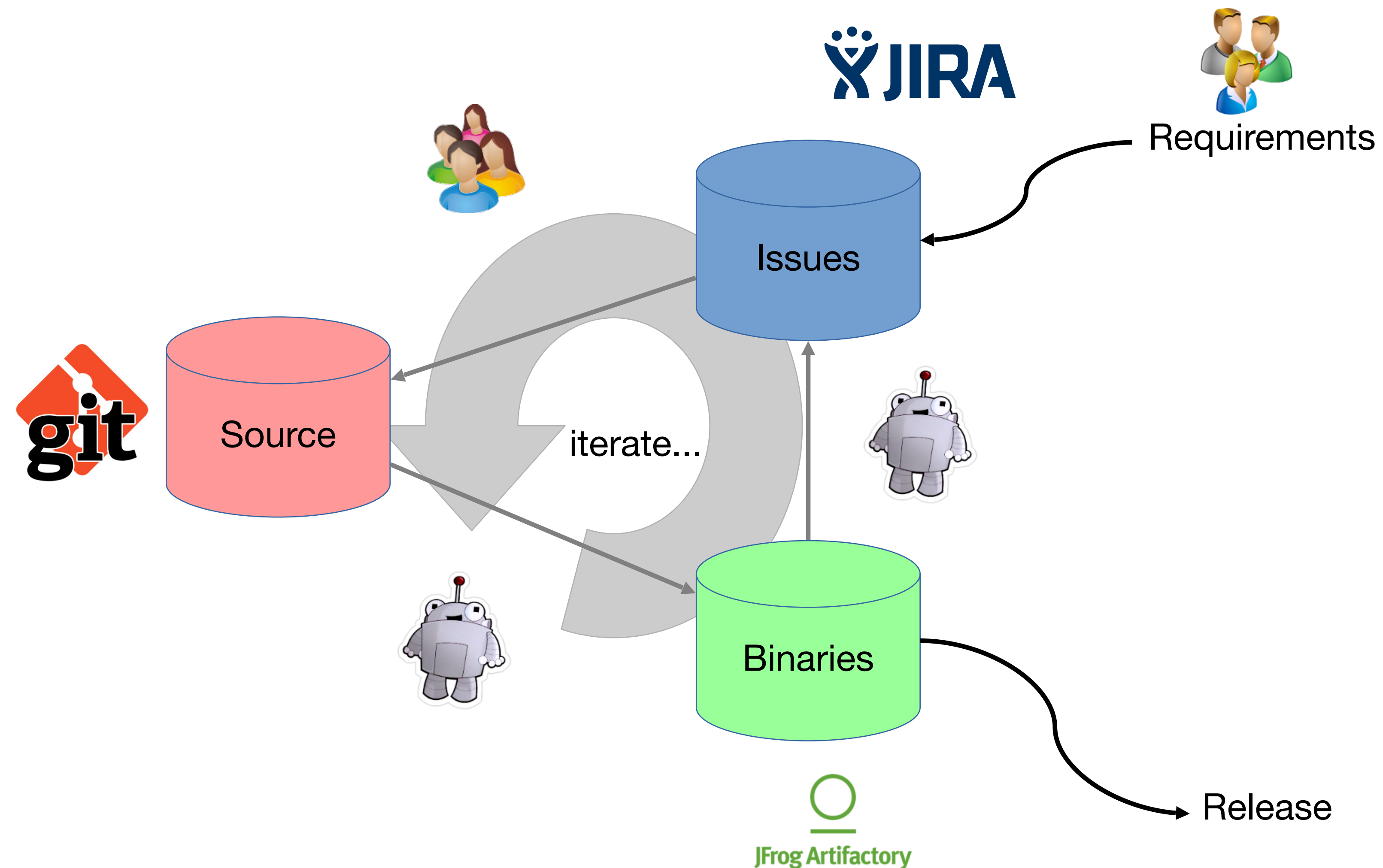


● Monitoring and Benchmarking

- Finding anomalies (performance, failures, ...) in complex applications and infrastructures

Yet another example...

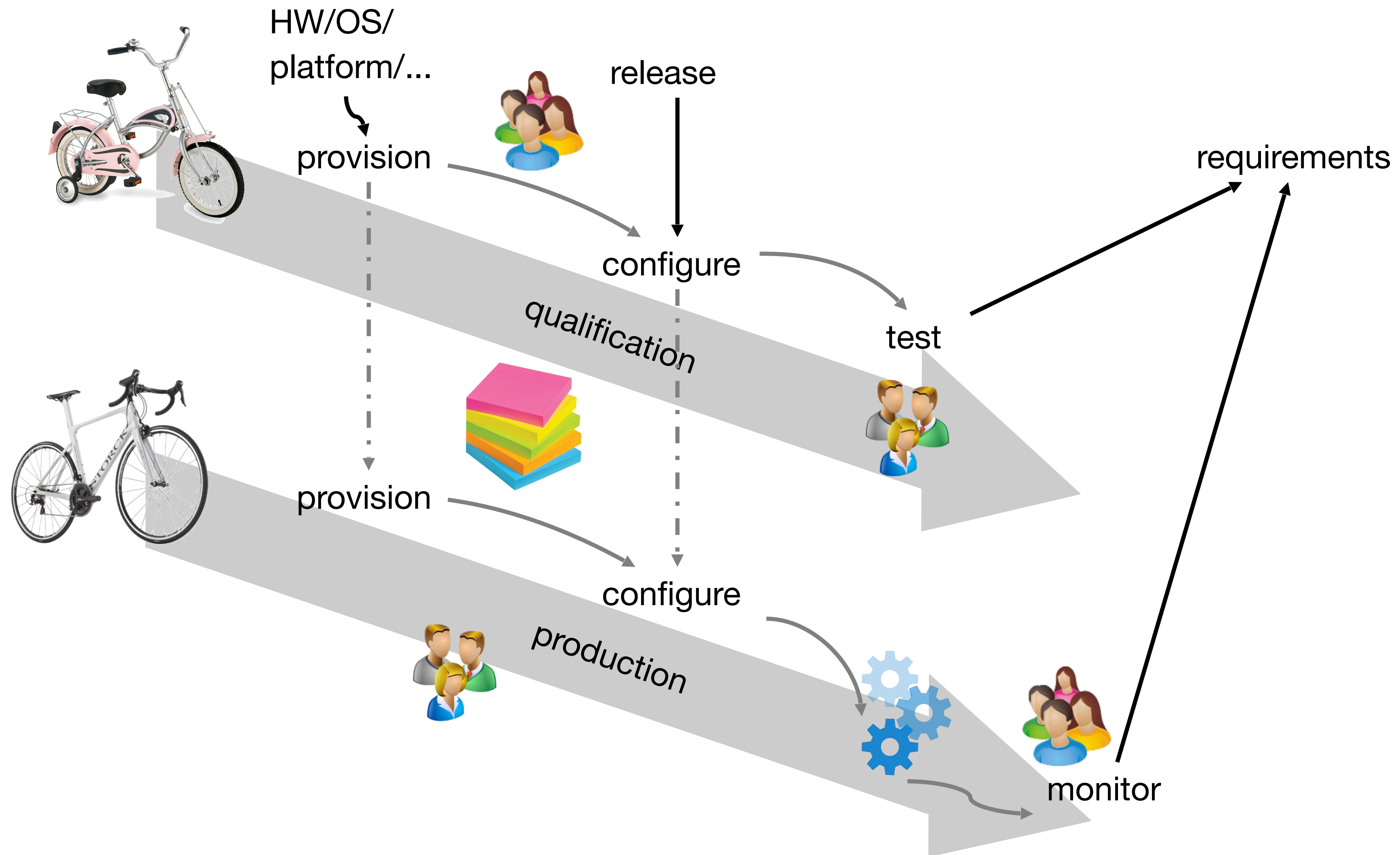
Agile development



Agile development

- ◎ Rests on formal (versioned) source, binary and documentation repositories
 - Unambiguous current state
 - Ability to back track
- ◎ Automated build and test
 - Fast feedback
- ◎ Fast iterations
- ◎ **Quick reaction to frequent small changes!**

Operations

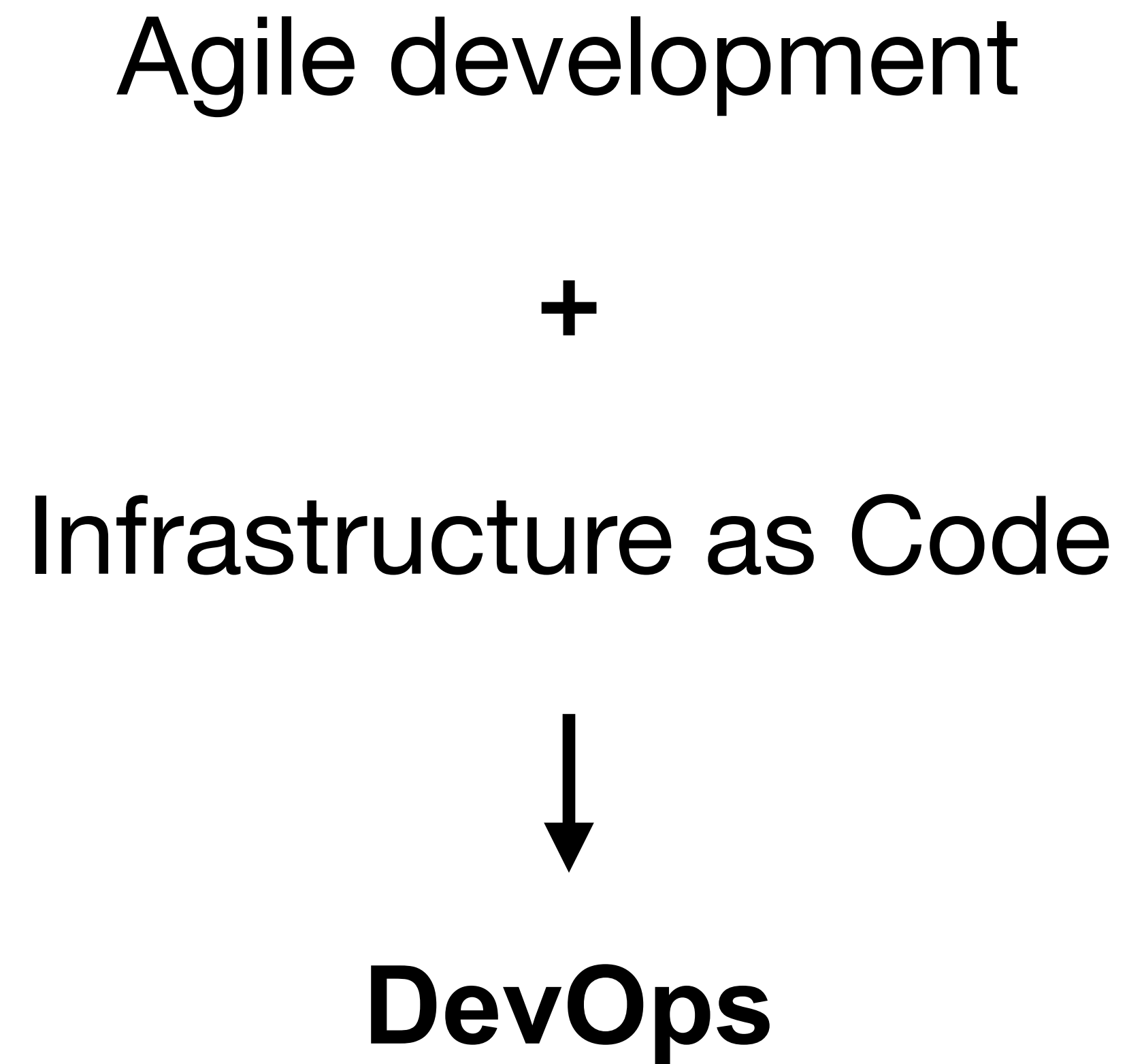


Operations

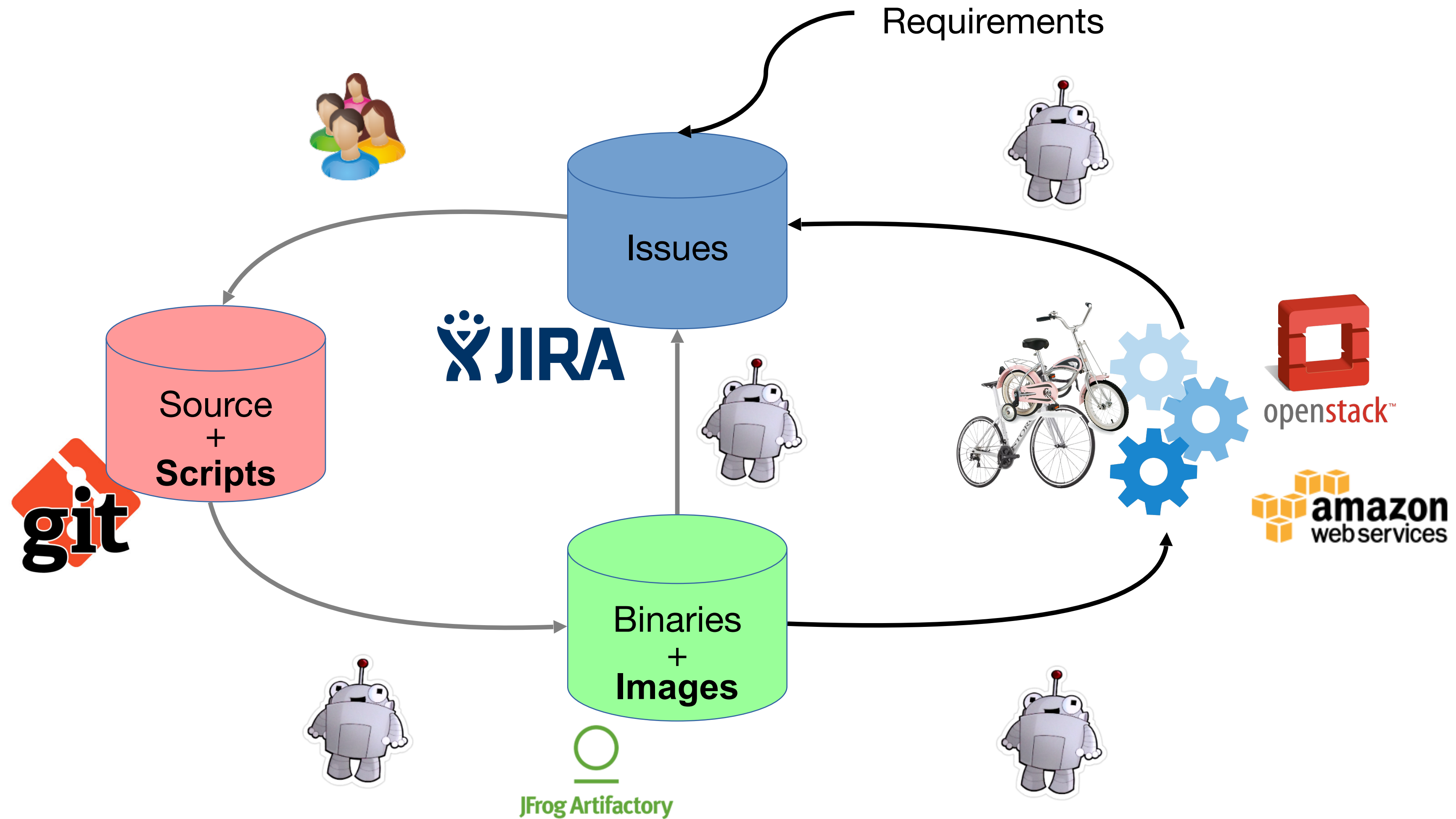
- Manual provisioning and configuration
- Informal communication between qualification and production stages
- Consequences:
 - Not repeatable or reproducible
 - Configuration drift leading to “Snowflake” servers (don't even look at it...)
 - Subjective monitoring

Infrastructure as Code

- Hardware provisioning with scripts
 - In contrast to: physically unboxing, plugging and configuring
- Software provisioning with scripts
 - In contrast to: clicking through setup wizards
- Configuration with scripts
 - In contrast to: clicking through control panels
- Include infrastructure scripts in agile process!



DevOps



DevOps

- Development and deployment are self-documenting and versioned
- Deployment is reproducible and repeatable
- Servers are disposable and consistent
- Supports fast, frequent, small changes!

Roadmap

- Distributed applications architectures & deployment
 - Common distribution patterns
 - Reliability and scalability
 - Automatic provision and configuration
- Cloud computing
 - Services and abstractions
 - Infrastructure management
 - virtualization, storage, ...
- Monitoring and evaluation
 - Monitoring frameworks and metrics
 - Benchmarking

Assessment

- Project (50%) – minimum grade: 10 values
 - Checkpoint #1: 08/11/2026
 - Checkpoint #2: 06/12/2026
 - Final Report and Submission: 30/12/2026
 - Presentation: 11-16/01/2027
- Written exam (50%) – minimum grade: 8 values
 - 04/01/2027

Project

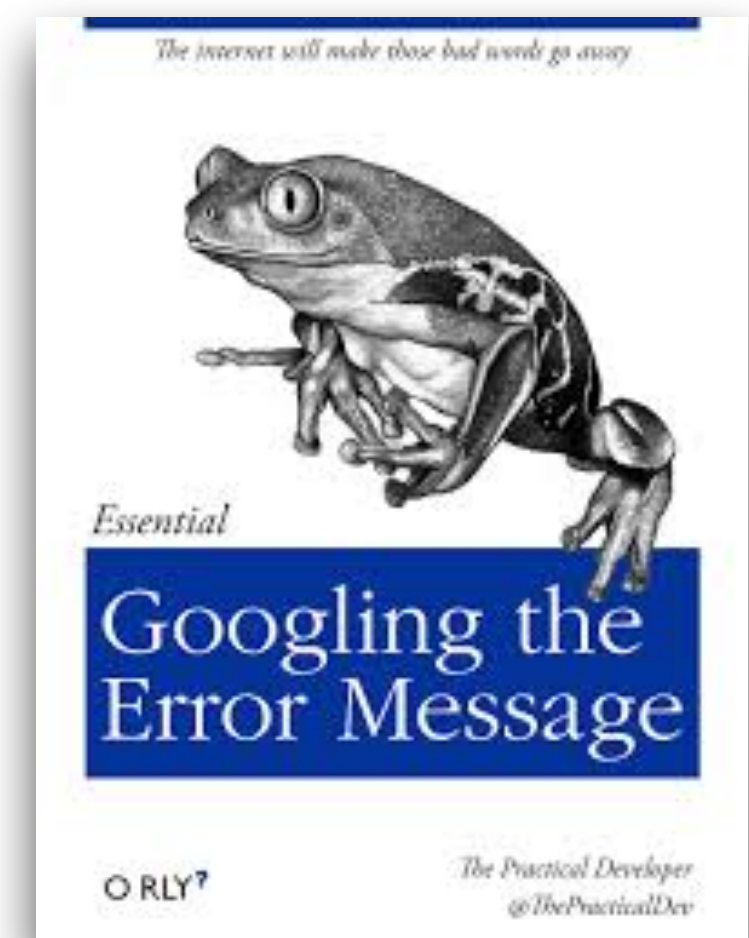
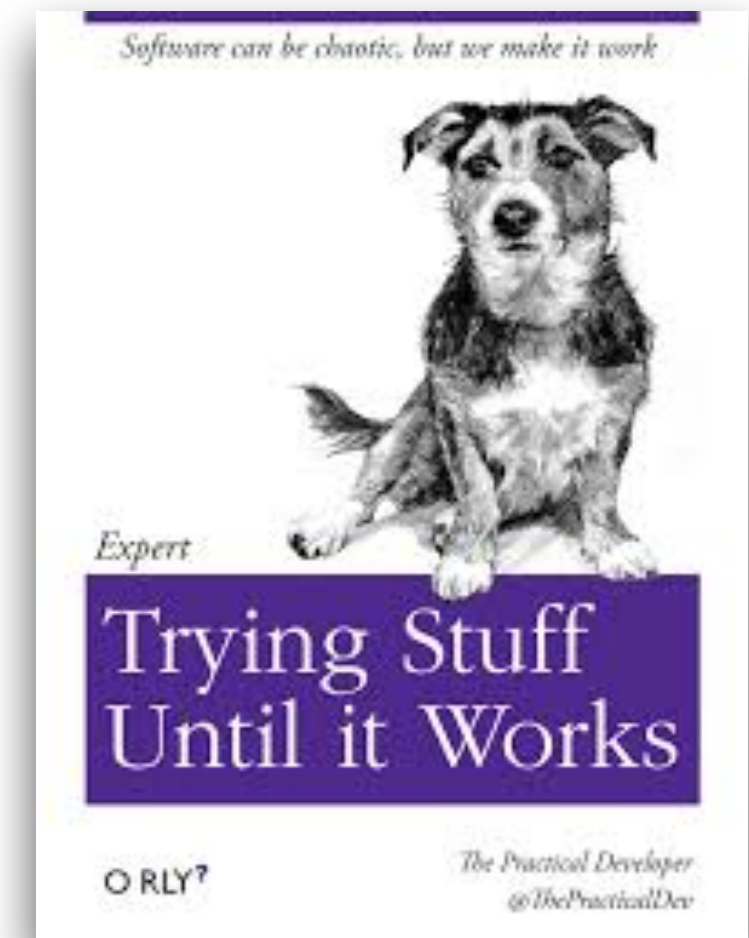
- Invite to the ASCN Platform and GitHub organization
 - An **invitation to the platform is going to be sent by email** (from **Fellowly**)
 - Institutional e-mail, or the email used in your application (for students who still don't have the institutional one)
 - After accepting the platform invitation, we will use the provided information to add you to the GitHub organization (make sure you have a GitHub account)
- ASCN Github organization
 - When you are granted access, **carefully read** the “How to use the ASCN-UM organization” at the **Instructions discussion thread**
- Each group should have **5 elements**
 - Start talking with your colleagues to find a group!
 - The organization has a **Groups discussion thread** to find members/groups

ASCN Website

- ASCN resources (slides, guides, bibliography, etc) will be provided on the website
- <https://distributedum.github.io/ascn/>

Main References

- K. Morris. *Infrastructure as Code: Managing Servers in the Cloud*. O'Reilly, 2016
- M. Kleppmann. *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*. O'Reilly, 2017
- T. Erl, R. Puttini e Z. Mahmood. *Cloud Computing: Concepts, Technology and Architecture*. Prentice Hall, 2013
- R. Jain. *The art of computer systems performance analysis - techniques for experimental design, measurement, simulation, and modeling*. Wiley, 1991



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Questions?